

Arithmetic Mean

Example:

- Find the arithmetic mean of the following frequency distribution:

x	1	2	3	4	5	6	7
f	5	9	12	17	14	10	6

Solution:

The arithmetic mean ($\bar{x}$) for a discrete frequency distribution is calculated using the formula:

$$\bar{x} = \frac{\sum(f \cdot x)}{\sum f}$$

First, we calculate the sum of frequencies ($\sum f$) and the sum of the product of the value and its frequency ($\sum(f \cdot x)$):

x	f	$f \cdot x$
1	5	5
2	9	18
3	12	36
4	17	68
5	14	70
6	10	60
7	6	42
Total	$\sum f = 73$	$\sum(f \cdot x) = 299$

Now, we substitute the values into the formula:

$$\bar{x} = \frac{299}{73} \approx \mathbf{4.0959}$$

Example:

- Calculate the mean for the following frequency distribution:

Class interval	0-8	8-16	16-24	24-32	32-40	40-48
Frequency	8	7	16	24	15	7

Solution:

The arithmetic mean ($\bar{x}$) for a grouped frequency distribution is calculated using the formula:

$$\bar{x} = \frac{\sum(f \cdot m)}{\sum f}$$

where m is the **class mark** (midpoint) of the class interval.

First, we find the class mark (m) for each interval, calculate $f \cdot m$, and then find the sums.

Class interval	Frequency (f)	Midpoint (m)	$f \cdot m$
0-8	8	4	32
8-16	7	12	84
16-24	16	20	320
24-32	24	28	672
32-40	15	36	540
40-48	7	44	308
Total	$\sum f = 77$		$\sum(f \cdot m) = 1956$

Now, we substitute the values into the formula:

$$\bar{x} = \frac{1956}{77} \approx 25.4026$$

Example:

- The average salary of male employees in a firm was Rs. 5,200 and that of females was Rs. 4,200. The mean salary of all the employees was Rs. 5,000. Find the percentage of male and female employees.

Solution:

This problem uses the concept of the **combined mean** ($\bar{x}_{12}$), which is defined as:

$$\bar{x}_{12} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$$

Let n_1 be the number of male employees and n_2 be the number of female employees. Let p_1 be the proportion of male employees, $p_1 = \frac{n_1}{n_1+n_2}$, and p_2 be the proportion of female employees, $p_2 = \frac{n_2}{n_1+n_2}$. Note that $p_1 + p_2 = 1$.

The formula can be rewritten in terms of proportions:

$$\bar{x}_{12} = p_1\bar{x}_1 + p_2\bar{x}_2$$

Given:

- $\bar{x}_{12} = 5000$
- $\bar{x}_1 = 5200$ (Male)
- $\bar{x}_2 = 4200$ (Female)
- $p_2 = 1 - p_1$

Substitute the values into the equation:

$$5000 = p_1(5200) + (1 - p_1)(4200)$$

$$5000 = 5200p_1 + 4200 - 4200p_1$$

$$5000 - 4200 = 5200p_1 - 4200p_1$$

$$800 = 1000p_1$$

$$p_1 = \frac{800}{1000} = 0.8$$

The proportion of male employees is $p_1 = 0.8$, so the percentage of male employees is $0.8 \times 100 = 80\%$.

The proportion of female employees is $p_2 = 1 - p_1 = 1 - 0.8 = 0.2$, so the percentage of female employees is $0.2 \times 100 = 20\%$.